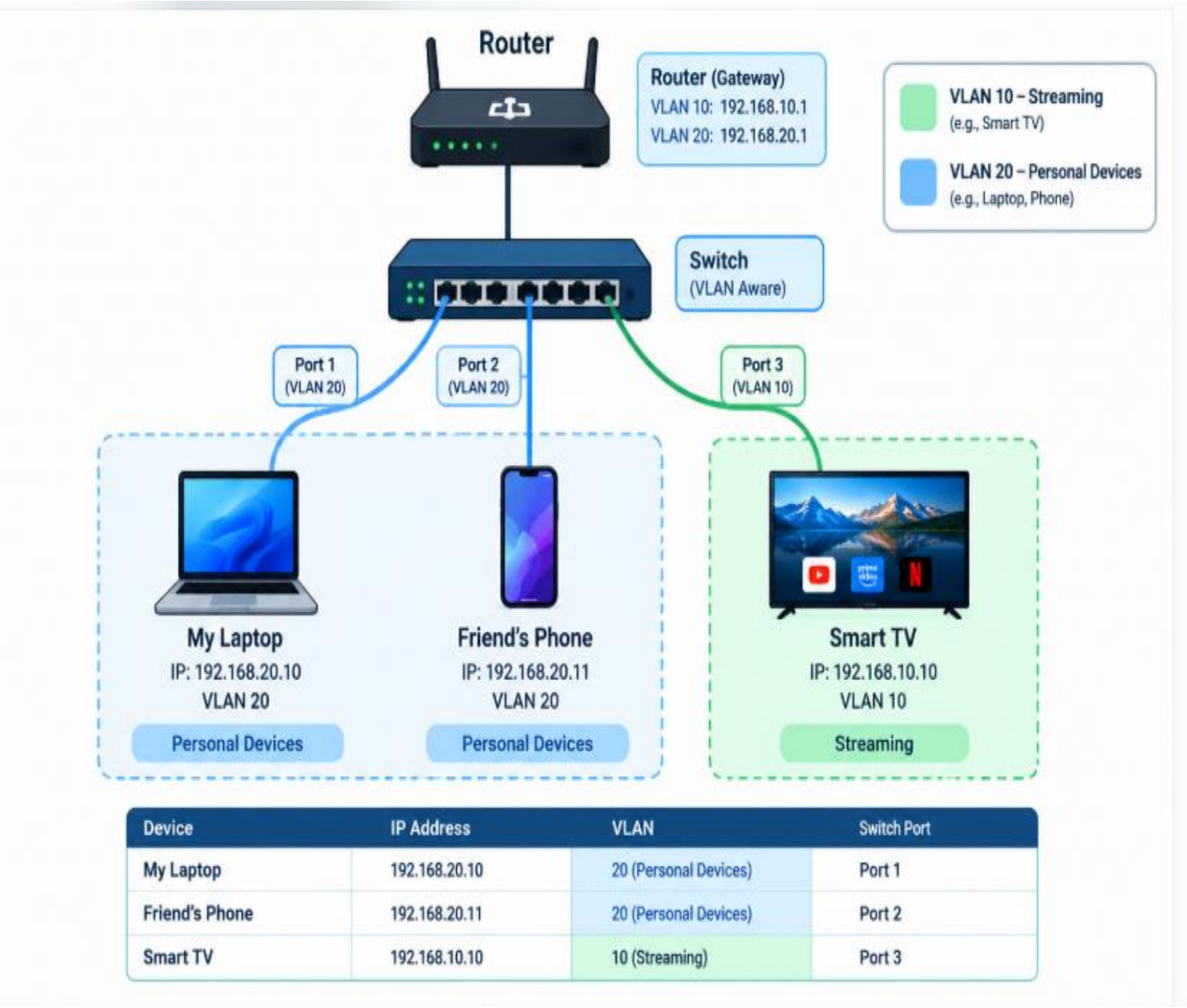
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Assignment-10

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1. **Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).**



Device	IP Address	VLAN	Switch Port
My Laptop	192.168.20.10	20 (Personal Devices)	Port 1
Friend's Phone	192.168.20.11	20 (Personal Devices)	Port 2
Smart TV	192.168.10.10	10 (Streaming)	Port 3

2. **Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.**

Step 1: Create the Topology

- **Add 1 Switch (2960) and 5 PCs in Cisco Packet Tracer.**
- **Connect all PCs to the switch using Copper Straight-Through cables.**

Step 2: Assign IP Addresses

PC	IP Address	Subnet Mask	VLAN
PC0	192.168.10.1	255.255.255.0	10
PC1	192.168.10.2	255.255.255.0	10
PC2	192.168.10.3	255.255.255.0	10
PC3	192.168.20.1	255.255.255.0	20
PC4	192.168.20.2	255.255.255.0	20

Enter the IP addresses under PC → Desktop → IP Configuration.

Step 3: Configure the Switch

Open the switch CLI and enter these commands:

```
enable
configure terminal

vlan 10
name VLAN10
exit

vlan 20
name VLAN20
exit

interface range fa0/1-3
switchport mode access
switchport access vlan 10
exit

interface range fa0/4-5
switchport mode access
```

```
switchport access vlan 20
exit


end
```

- VLAN 10 contains PC0, PC1, and PC2.
- VLAN 20 contains PC3 and PC4.

Step 4: Verify the VLANs

Run this command:

```
show vlan brief
```

It displays the VLANs and their assigned switch ports.

Step 5: Test Ping

Test PCs in the same VLAN:

From PC0, run:

```
ping 192.168.10.2
```

Expected result: Ping successful.

Test PCs in different VLANs:

From PC0, run:

```
ping 192.168.20.1
```

Expected result: Ping fails.

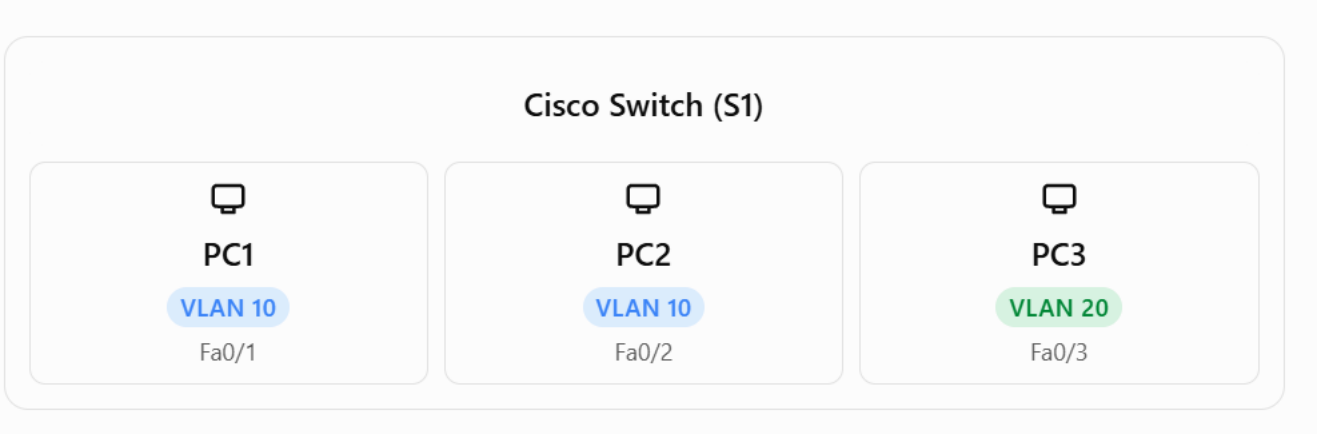
3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

VLAN 10 to VLAN 20 Configuration (Cisco Packet Tracer)

Aim: Move PC1 from VLAN 10 to VLAN 20 and test connectivity using the ping command.

1. Before Configuration

- PC1 and PC2 are in VLAN 10.
- PC3 is in VLAN 20.



2. Before Configuration Commands

Enter these commands in the switch CLI:

```
enable
configure terminal
vlan 10
name SALES
exit
vlan 20
name HR
exit
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit
interface fastEthernet 0/2
switchport mode access
switchport access vlan 10
exit
interface fastEthernet 0/3
switchport mode access
switchport access vlan 20
end
```

```
show vlan brief
```

PC1 is connected to port Fa0/1, which belongs to VLAN 10.

3. Move PC1 to VLAN 20

Only change the VLAN assigned to PC1's switch port.

```
enable
configure terminal
interface fastEthernet 0/1
switchport access vlan 20
end
show vlan brief
```

4. After Configuration

Switch port	Before	After
Fa0/1 (PC1)	VLAN 10	VLAN 20
Fa0/2 (PC2)	VLAN 10	VLAN 10
Fa0/3 (PC3)	VLAN 20	VLAN 20

PC1 has now moved from VLAN 10 to VLAN 20.

5. Test Connectivity

Assume the PCs have these IP addresses on the same subnet:

PC	IP address
PC1	192.168.1.1
PC2	192.168.1.2
PC3	192.168.1.3

Open each PC's Desktop → Command Prompt and run the following tests.

Before moving PC1:

```
PC1> ping 192.168.1.2
```

Expected result: Success, because PC1 and PC2 are in VLAN 10.

After moving PC1 to VLAN 20:

PC1> ping 192.168.1.2

Expected result: Fail, because PC1 and PC2 are now in different VLANs.

PC1> ping 192.168.1.3

Expected result: Success, because PC1 and PC3 are in VLAN 20.

Note: These expected results assume the PCs have correct IP configurations, their links are up, and no other settings block connectivity. The first ping may time out while ARP resolves.

4. **Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).Constraint:Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.**

Cisco Packet Tracer: Assign 10 Ports to VLAN 30

Commands:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 30
Switch(config-vlan)# exit
Switch(config)# interface range fastEthernet 0/1-10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# end
Switch# show vlan brief
```

Explanation:

- **vlan 30 — Creates VLAN 30.**

- **interface range fastEthernet 0/1-10 — Selects ports Fa0/1 to Fa0/10 together.**
- **switchport mode access — Sets the ports to access mode.**
- **switchport access vlan 30 — Assigns all 10 ports to VLAN 30.**
- **show vlan brief — Verifies the VLAN configuration.**

Expected result: Ports Fa0/1 through Fa0/10 appear under VLAN 30.